Lahore University of Management Sciences

AI 501 – Mathematics for Artificial Intelligence

Fall 2026

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Course URL (if any)	LMS

Course Teaching Methodology (Please mention following details in plain text)
• Teaching Methodology: In-person classes (2.5 hours meeting once a week) • Attendance is mandatory as maintaining a good attendance will help students in many ways. Students not frequently attending the lecture will find it difficult to cope with the course. We will also have assessments during the class.

Course Basics				
Credit Hours	3			
Lecture(s)	Nbr of Lec(s) Per Week	1	Duration	2 hours and 30 minutes
Recitation/Lab (per week)	Nbr of Lec(s) Per Week	0	Duration	
Tutorial (per week)	Nbr of Lec(s) Per Week	1	Duration	1 hour (if needed)

Course Distribution	
Core	MS AI
Elective	None
Open for Student Category	MS AI
Close for Student Category	All other students (Open for Auditing)

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COURSE DESCRIPTION
This course develops the mathematical foundations on which machine learning and artificial intelligence are built. It is the first course of the graduate programme and is designed for a cohort entering from varied backgrounds, including computer science, engineering, economics, and management. The primary objective is therefore to bring all students to a common and rigorous foundational level, so that subsequent courses in machine learning, deep learning, and related areas can proceed from a shared mathematical vocabulary. The treatment is deliberately sequential. Mathematical tools are developed in dependency order, and each block of theory concludes with a machine-learning method that the theory makes possible. Learning algorithms are introduced as anchors and worked examples rather than as a survey; the subject of the course remains the mathematics. Several central results are derived more than once by different routes, for example least squares obtained first by orthogonal projection and subsequently by differentiation, and logistic regression obtained first as a convex optimization problem and subsequently as maximum-likelihood estimation, since re-derivation is the mechanism through which a cohort of mixed preparation is consolidated. In broad brush terms, the following topics are covered. **Vectors, Vector Spaces, and Linear Transformations.** Representation of data as vectors; inner products, norms, distance, angle, and similarity measures; vector spaces and subspaces, linear independence, basis, and dimension; orthogonality, orthogonal projection, and Gram-Schmidt orthogonalization; matrices as linear maps, rank, column space and null space, and the geometric interpretation of matrix operations. **Linear Systems and Least Squares.** Formulation and solvability of systems of linear equations; matrix inverses and the Moore-Penrose pseudo-inverse; the least-squares problem derived geometrically; conditioning and numerical considerations in practical computation. **Matrix Decompositions.** Eigenvalue decomposition, the spectral theorem for symmetric matrices, quadratic forms, and positive semi-definiteness; singular value decomposition and its geometric interpretation; low-rank approximation and its role in compression, denoising, and data representation. **Dimensionality Reduction.** The curse of dimensionality; principal component analysis derived both as an eigen-decomposition of the empirical covariance matrix and as a singular value decomposition of the centred data matrix, together with component selection, whitening, and low-rank data representation. **Multivariate and Matrix Calculus.** Partial derivatives, gradients, and directional derivatives; the multivariate chain rule; Jacobian and Hessian matrices; matrix-calculus identities under an explicit layout convention; second-order Taylor expansion. The course closes by applying this machinery to the composition of linear maps and nonlinearities that constitutes a feed-forward neural network, identifying backpropagation as an application of the multivariate chain rule. **Optimization.** Optimality conditions and their relationship to the Hessian; convex sets and convex functions; gradient descent and its stochastic and mini-batch variants; Newton and quasi-Newton methods; constrained optimization, Lagrange multipliers, duality, and the Karush-Kuhn-Tucker conditions.

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Probability and Distributions. Axioms of probability, conditional probability, Bayes theorem, and independence; discrete and continuous random variables, expectation, and variance; standard univariate distributions; joint and conditional distributions, covariance, and the multivariate Gaussian; laws of large numbers and the central limit theorem.

Statistical Inference and Information. Maximum-likelihood and maximum a posteriori estimation; properties of estimators, including bias, variance, and consistency; entropy, cross-entropy, and Kullback-Leibler divergence, and the identification of cross-entropy minimization with maximum-likelihood estimation; the Bayesian framework, conjugate priors, posterior predictive distributions, and an overview of approximate inference and sampling methods.

Learning Methods Used as Anchors. Nearest-neighbour classification, linear and polynomial regression, ridge regression and regularization, logistic regression, support vector machines and the kernel trick, naive Bayes classification, and Bayesian linear regression. Model evaluation and selection are treated alongside these, covering train, validation, and test partitions, cross-validation, classification metrics, and the bias-variance decomposition.

Computational Practice. Each session includes a short demonstration in which the mathematics of that session is implemented numerically, with attention to vectorization, numerical conditioning, and the divergence between analytical results and finite-precision computation. This material is illustrative and supports understanding of the theory rather than forming a separately assessed component.

COURSE PREREQUISITE(S)	
•	Pre-requisites: None
•	Co-requisites: None

COURSE OBJECTIVES
• Foundational Knowledge: Equip students with a robust understanding of linear algebra, calculus, probability theory, and optimization techniques, which are essential for ML and AI. • Practical Applications: Demonstrate the application of mathematical concepts in real-world ML and AI scenarios, including data processing, model training, and performance evaluation. • Analytical Thinking: Foster analytical thinking by encouraging students to critically assess different algorithms and their suitability for various tasks, considering both theoretical foundations and

	practical applications.

Grading Distribution: Component Details and weightages
Quizzes, 25 % CP, 5% Mid-Exam and Mid-Viva, 30 % Final Exam and Final Viva, 40 %

Textbook(s)/Supplementary Readings
Books: • S.Boyd and L. Vandenberghe. **Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares.** Cambridge University Press, 2019 • M. P. Deisenroth, A. A. Faisal and Cheng Soon Ong. **Mathematics for Machine Learning**. Cambridge University Press, 2019 • Ali H. Sayed, **Inference and Learning from Data,** Foundations (Volume 1). Cambridge University Press, 2022 • G. Strang. **Introduction to Linear Algebra.** 2016 • J. A. Gubner, **Probability and Random Processes for Electrical and Computer Engineers**, Cambridge University Press, 2006. • S. L. Miller and D. Childers, **Probability and Random Processes: With Applications to Signal Processing and Communications.** • A. Papoulis and S.U. Pillai, **Probability, Random Variables, and Stochastic Processes.** • Class notes will be provided to supplement these readings

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Support Services
LUMS offers a range of academic and other services to support students. These are mentioned below, and you are encouraged to use these in addition to in-class assistance from course staff. For a complete list of campus support services available for you click here (https://advising.lums.edu.pk/#supportservices).

Campus supports & Key university policies
Campus Supports Students are strongly encouraged to meet course instructors and TA's during office hours for assistance in course-content, understand the course's expectations from enrolled students, etc. Beyond the course, students are also encouraged to use a variety of other resources. (Instructors are also encouraged to refer students to these resources when needed.) These resources include Counseling and Psychological Services/CAPS (for mental health), LUMS Medical Center/LMC (for physical health), Office of Accessibility & Inclusion/ OAI (for long-term disabilities), advising staff dedicated to supporting and guiding students in each school, online resources (https://advising.lums.edu.pk/advising-resources), etc. To view all support services, their specific role as well as contact information click here (https://advising.lums.edu.pk/#supportservices). **Academic Honesty/Plagiarism** LUMS has zero tolerance for academic dishonesty. Students are responsible for upholding academic integrity. If unsure, refer to the student handbook and consult with instructors/teaching assistants. To check for plagiarism before essay submission, use similarity@lums.edu.pk. Consult the following resources: 1) Academic and Intellectual Integrity (http://surl.li/gpvwb), and 2) Understanding and Avoiding Plagiarism (http://surl.li/gpvwo). **LUMS Academic Accommodations/ Petitions policy** Long-term medical conditions are accommodated through the Office of Accessibility & Inclusion (OAI). Short-term emergencies that impact studies are either handled by the course instructor or Student Support Services (SSS). For more information, please see Missed Instrument or 'Petition' FAQs for students and faculty (https://rb.gy/8sj1h) **LUMS Sexual Harassment Policy** LUMS and this class are a harassment-free zone. No behavior that makes someone uncomfortable or negatively impacts the class or individual's potential will be tolerated. To report sexual harassment experienced or observed in class, please contact me. For further support or to file a complaint, contact OAI at oai@lums.edu.pk or harassment@lums.edu.pk. You may choose to file an informal or formal complaint to put an end to the offending behavior. You can also call their Anti-Harassment helpline at 042-35608877 for advice or concerns. *For more information: Harassment, Bullying & Other Interpersonal Misconduct: Presentation (http://surl.li/gpvwt)*

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Course Topics and Schedule

Week 1: Vectors, Geometry, and Similarity

- Course Overview: objectives, structure, dependency map of the term, notation conventions
- Data as Vectors: feature vectors, one-hot encodings, images and documents as vectors
- Vector Operations: addition, scaling, linear combination, affine and convex combinations
- Inner Product and Norms: L1, L2, and Lp norms, the L0 counting function
- Distance, Angle, and Cauchy-Schwarz Inequality
- Similarity Measures: cosine similarity, correlation coefficient as cosine similarity of centred vectors

Week 2: Vector Spaces, Orthogonality, and Projection

- Vector Spaces and Subspaces: axioms and examples, including polynomial and function spaces
- Span, Linear Independence, Basis, and Dimension
- Coordinates and Change of Basis
- Orthogonality and Orthonormal Bases
- Orthogonal Projection: onto a line and onto a subspace
- Gram-Schmidt Orthogonalization and QR Decomposition

Week 3: Matrices as Linear Transformations

- Matrix Notation and Operations: addition, multiplication, transpose, trace
- Matrix-Vector Product: three interpretations as row-wise inner products, combination of columns, and action of a linear map
- Matrix-Matrix Product, Outer Products, and Block Matrices
- Special Matrices: diagonal, symmetric, orthogonal, projection, permutation
- Column Space, Null Space, Rank, and the Rank-Nullity Theorem
- Geometric Transformations: rotation, scaling, shear, reflection, projection

Week 4: Linear Systems, Least Squares, and the Supervised Learning Frame

- Systems of Linear Equations: existence and uniqueness characterized through rank, over-determined and under-determined cases
- Gaussian Elimination and LU Decomposition (brief)
- Matrix Inverse: conditions for existence, conditioning, and computational caution against explicit inversion
- Determinant: geometric interpretation as volume scaling
- Least Squares by Orthogonal Projection: derivation of the normal equations without recourse to calculus
- Moore-Penrose Pseudo-Inverse and the Minimum-Norm Solution
- Supervised Learning Nomenclature: data, features, labels, hypothesis, loss
- Linear Regression as the First Learning Algorithm

Week 5: Model Selection, Evaluation, and Regularization

- k-Nearest-Neighbour Classification and Regression: distance metrics revisited, decision boundaries, choice of k
- Classifier Performance Evaluation: confusion matrix, accuracy, precision, recall, F1-score, ROC, AUC, class imbalance
- Train, Validation, and Test Partitions; k-Fold Cross-Validation

- Polynomial Feature Expansion, Underfitting and Overfitting
- Bias-Variance Decomposition
- Ridge Regression and Regularization: modification of the normal equations and the conditioning argument

Week 6: Eigenvalue and Singular Value Decompositions, and Dimensionality Reduction

- Eigenvalues and Eigenvectors: characteristic equation, diagonalization
- Spectral Theorem for Symmetric Matrices
- Quadratic Forms, Positive Definiteness and Semi-Definiteness, and their Eigenvalue Characterization
- Singular Value Decomposition: statement, geometric interpretation, relationship to eigenvalue decomposition, pseudo-inverse via SVD
- Low-Rank Approximation and the Eckart-Young Theorem; matrix norms
- Curse of Dimensionality
- Principal Component Analysis: derived as eigen-decomposition of the empirical covariance matrix and as SVD of the centred data matrix; component selection and whitening

Week 7: Mid-Exam

- Covering Weeks 1 to 6

Week 8: Multivariate and Matrix Calculus

- Functions of Several Variables: limits and continuity
- Partial Derivatives, Directional Derivatives, and the Gradient
- The Multivariate Chain Rule
- Jacobian and Hessian Matrices
- Matrix Calculus: explicit layout convention, gradients of linear and quadratic forms
- Second-Order Taylor Expansion
- Application: analytical re-derivation of the least-squares solution and its agreement with the Week 4 projection result

Week 9: Unconstrained Optimization

- Objective Functions, Stationary Points, Local and Global Minima
- First-Order and Second-Order Optimality Conditions via the Gradient and Hessian
- Convex Sets and Convex Functions: definitions, the Hessian test, operations preserving convexity
- Gradient Descent: algorithm, step-size selection, convergence behaviour, role of conditioning
- Stochastic and Mini-Batch Gradient Descent
- Newton's Method, with brief treatment of quasi-Newton and conjugate gradient methods
- Logistic Regression as a Convex Optimization Problem: sigmoid, decision boundaries, loss function, gradient derivation, multi-class extension

Week 10: Constrained Optimization and Support Vector Machines

- Constrained Optimization, the Lagrangian, and Lagrange Multipliers
- The Dual Problem: weak and strong duality
- Karush-Kuhn-Tucker Conditions
- Hard-Margin Support Vector Machine: margin, primal and dual formulations
- Soft-Margin Support Vector Machine and the Hinge Loss
- Support Vectors interpreted through Complementary Slackness
- The Kernel Trick: common kernels and the positive semi-definiteness requirement

Week 11: Probability Theory Fundamentals

- Sample Spaces, Events, and the Axioms of Probability
- Conditional Probability, Law of Total Probability, Bayes Theorem
- Independence and Combinatorics
- Discrete Random Variables and Probability Mass Functions
- Continuous Random Variables and Probability Density Functions; Cumulative Distribution Functions
- Expectation, Variance, and Higher Moments
- Functions of a Random Variable and the Change-of-Variables Formula

Week 12: Distributions, Joint Behaviour, and Asymptotics

- Standard Distributions: Bernoulli, binomial, geometric, Poisson, uniform, exponential, Gaussian
- Joint, Marginal, and Conditional Distributions
- Covariance and Correlation, reconnected to the empirical covariance matrix of Week 6
- The Multivariate Gaussian: density, covariance as a positive semi-definite matrix, ellipsoidal level sets, eigen-decomposition of the covariance and the probabilistic reading of principal components
- Law of Large Numbers and the Central Limit Theorem

Week 13: Estimation and Information

- Maximum-Likelihood Estimation: likelihood, log-likelihood, worked derivations for Bernoulli and Gaussian parameters
- Properties of Estimators: bias, variance, consistency, and the relationship to the bias-variance decomposition
- Maximum a Posteriori Estimation: priors and the correspondence between MAP estimation and regularized least squares
- Entropy, Cross-Entropy, and Kullback-Leibler Divergence
- Cross-Entropy Minimization identified with Maximum-Likelihood Estimation, re-deriving the logistic regression loss probabilistically
- Naive Bayes Classifier, including text classification

Week 14: Bayesian Inference, Neural Networks, and Synthesis

- The Bayesian Framework: prior, likelihood, posterior, evidence
- Conjugate Priors: Beta-Bernoulli and Gaussian-Gaussian cases
- Bayesian Linear Regression and the Posterior Predictive Distribution
- Computational Challenges of the Evidence Integral
- Approximate Inference and Sampling (overview): Laplace approximation, rejection and importance sampling, Monte Carlo integration including the Box-Muller transform, Markov Chain Monte Carlo, variational inference
- Feed-Forward Neural Networks as compositions of linear maps and nonlinearities; backpropagation as an application of the multivariate chain rule
- Course Synthesis: mapping each mathematical construct introduced during the term to its role in machine learning practice